








# Blockchain Applications

## Lecture 2: Stylized Facts, Efficiency and Bubbles



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# Opening: How Do Crypto Prices Behave?



Eugene F. Fama. Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons

*"A market in which prices always 'fully reflect' available information is called 'efficient.'"*
— *Fama (1970)*

- Bitcoin's annualised volatility in 2018–2026 is 64.1%, against 19.2% for the S&P 500
- Crashes: Bitcoin had 17 days beyond 4 standard deviations in 3 183 days, against 0.2 expected under the Normal distribution
  - worst day, 12 March 2020: a log return of $-46.5\%$
- Volatility clusters: the Ljung–Box statistic $Q(10)$ is 212.7 for squared returns, against 20.2 for returns
- Robust variance-ratio tests do not reject a random walk at 2 to 10 days in any sub-period
  - not rejecting a random walk is not proof of efficiency
- The log-periodic power law (LPPL) fits the 2017 bubble with $R^2 \approx 0.98$, yet the estimated crash date moves with the window

▸ Central chart: Efficiency Through Time

## Today's Question and Route

- **Question**: how do crypto prices behave, and what does that mean for a business that holds, accepts or prices crypto?
  - a treasurer, an exchange or a fund needs numbers, not narratives
- **Route**
  - from prices to returns on a 7-day market: Lecture 1 described levels and trends; today the same prices become returns
  - the shape of returns: heavy tails, rare but frequent crashes
  - volatility through time, and the link with equities and gold
  - is the market efficient? can past prices predict returns?
  - bubbles, drawdowns and network value (Metcalfe's law)
- Lecture 4: models for volatility and tail risk: generalised autoregressive conditional heteroskedasticity (GARCH), value at risk (VaR), expected shortfall (ES)

## Learning Outcomes

- Compute log returns and annualise volatility with the right number of days
- Read a summary table: volatility, skewness, kurtosis, worst day
- Use a histogram and a quantile–quantile (QQ) plot to show why the Normal distribution understates crash risk
- Interpret rolling volatility and rolling correlation charts
- Explain weak-form efficiency and read a variance-ratio test
- Measure drawdowns and judge bubble and network-value claims with data

# Abbreviations

**ACF** autocorrelation function
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**CI** confidence interval
**COVID-19** coronavirus disease 2019
**CRIX** CRypto IndeX (Trimborn & Härdle (2018))
**EODHD** EOD (end-of-day) Historical Data, data vendor
**ES** expected shortfall
**EWMA** exponentially weighted moving average
**FABIZ** Faculty of Business Administration in Foreign Languages, ASE
**FTX** crypto exchange (company name), bankrupt in November 2022
**GARCH** generalised autoregressive conditional heteroskedasticity
**i.i.d.** independent and identically distributed
**IDA** Institute Digital Assets
**JB** Jarque–Bera (test statistic)
**LPPL** log-periodic power law
**LPPLS** log-periodic power law singularity
**MBB** Master Blockchain for Business
**QQ** quantile–quantile
**S&P 500** Standard & Poor's 500 US stock index
**sd** standard deviation
**SE** standard error
**SFM** Statistics of Financial Markets (master course; Quantlet prefix)
**USD** US dollar
**UTC** Coordinated Universal Time
**VaR** value at risk
**VR** variance ratio

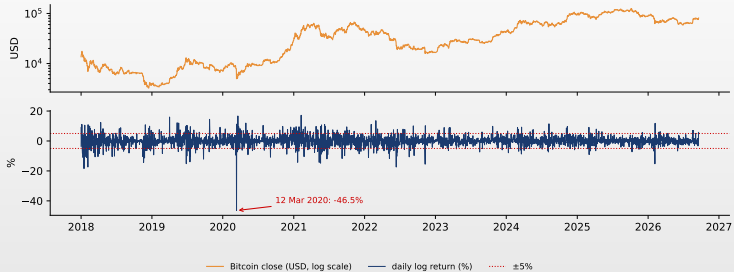
# Reading and Tools

- Core reading: Trimborn & Härdle (2018), *CRIX: an index for cryptocurrencies*; Peterson (2018), *Metcalfe's law as a model for Bitcoin's value*
- Video units on Quantinar (course Blockchain Analytics)
  - Blockchain Mechanism and Distributional Characteristics of Cryptos
  - Crypto Currency Returns
- Data
  - daily closes of Bitcoin, Ethereum, the S&P 500 and gold, 2018 – 18 September 2026 (EODHD)
  - on-chain transactions of Bitcoin, 2014–2025 (ETH Chronicles project, Lecture 1)
- Further: Pele et al. (2023), *Are cryptos becoming alternative assets?*

## Log Returns and Annualisation

- Daily **log return** in %: $r_t = 100\,(\ln P_t - \ln P_{t-1})$; $P_t$: the close on day $t$
- Annual volatility from daily data: $\sigma_a = s\sqrt{A}$
  - $s$: the standard deviation of daily returns; $A$: the number of returns per year
  - crypto trades every day: $A = 365$; stocks: $A \approx 252$ trading days
- **Worked example**: Bitcoin 2018–2026, $s = 3.355\%$ a day
  - right: $\sqrt{365} \times 3.355 = 19.10 \times 3.355 = 64.1\%$ a year
  - wrong: $\sqrt{252} \times 3.355 = 53.3\%$, an understatement of 17%
- Rule: annualise each asset with its own calendar; compare assets only after that

# From Prices to Returns: Bitcoin, 2018–2026




- Top: daily close in USD, log scale; bottom: daily log returns in %, 1 January 2018 – 18 September 2026, $n = 3\,183$
- The price trends and swings; the returns stay around zero, with bursts: 340 days (10.7%) beyond $\pm 5\%$
- Worst day 12 March 2020: $-46.5\%$; best day 8 February 2021: $+17.2\%$
- Log returns add up: the sum of the 3 183 daily returns, 174.3, equals $100 \ln(P_{\text{end}}/P_{\text{start}})$

BA_L2_returns_path

## Moments of the Return Distribution

- For $n$ daily returns $r_1, \dots, r_n$:

$$\bar{r} = \frac{1}{n}\sum_{t=1}^{n} r_t, \qquad s^2 = \frac{1}{n-1}\sum_{t=1}^{n}(r_t - \bar{r})^2$$

$$\hat{S} = \frac{1}{n}\sum_{t=1}^{n}\left(\frac{r_t - \bar{r}}{s}\right)^3, \qquad \hat{K} = \frac{1}{n}\sum_{t=1}^{n}\left(\frac{r_t - \bar{r}}{s}\right)^4 - 3$$

  - $\bar{r}$: mean; $s$: standard deviation (volatility); $\hat{S}$: skewness; $\hat{K}$: excess kurtosis
  - Normal distribution: $S = 0$ and $K = 0$

- **Jarque–Bera (JB) test** of normality:

$$\mathrm{JB} = \frac{n}{6}\left(\hat{S}^2 + \frac{\hat{K}^2}{4}\right) \sim \chi^2_2 \text{ under } H_0\text{: Normal}$$

  - reject normality at 5% if $\mathrm{JB} > 5.99$
  - 2018–2026: Bitcoin 29 620, Ethereum 16 593, S&P 500 20 462, gold 10 041: normality is rejected for all four

## Worked Example: Jarque–Bera for Bitcoin

- Bitcoin, 2018–2026: $n = 3\,183$, $\hat{S} = -0.945$, $\hat{K} = 14.825$

$$\mathrm{JB} = \frac{3\,183}{6}\left(0.945^2 + \frac{14.825^2}{4}\right) = 530.5 \times (0.893 + 54.945) \approx 29\,620$$

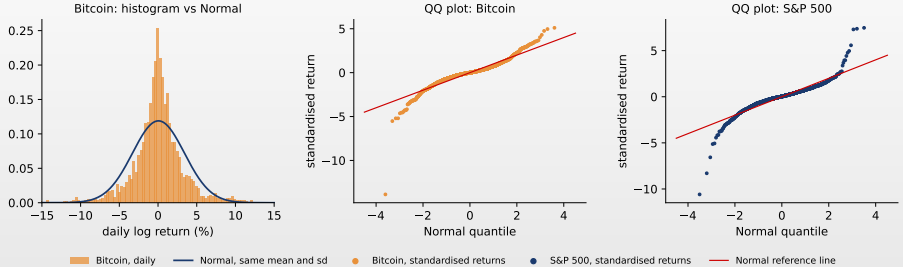
- 5% critical value of $\chi^2_2$: 5.99; p-value $< 10^{-300}$
- Almost all of the statistic comes from kurtosis (54.9 of 55.8): the problem is the tails, not the asymmetry
- Caution: with $n$ in the thousands, JB rejects even mild deviations; the size of $\hat{K}$ matters more than the p-value

# Four Assets Side by Side, 2018–2026

| | $n$ | sd (%) | ann. vol. (%) | skewness | excess kurtosis | worst day (log %) | days $\lvert r\rvert > 5\%$ (%) |
|---|---|---|---|---|---|---|---|
| Bitcoin | 3 183 | 3.36 | 64.1 | −0.95 | 14.8 | −46.5 | 10.7 |
| Ethereum | 3 183 | 4.41 | 84.3 | −0.83 | 11.1 | −55.1 | 17.8 |
| S&P 500 | 2 189 | 1.21 | 19.2 | −0.65 | 14.9 | −12.8 | 0.6 |
| Gold | 2 268 | 1.07 | 16.9 | −0.55 | 10.2 | −11.1 | 0.3 |

- Daily log returns, 1 January 2018 – 18 September 2026, each asset on its own calendar
  - crypto: every day, $A = 365$; S&P 500 and gold: Monday–Friday closes, $A = 252$
  - **skewness**: 0 for a symmetric distribution; negative: the left tail (losses) is longer
  - **excess kurtosis**: 0 for the Normal distribution; positive: more extreme days than the Normal predicts
- Interpretation
  - Bitcoin is 3.3 times as volatile as the S&P 500; it moves by more than 5% on one day in nine
  - the worst day of Bitcoin and Ethereum was the same: 12 March 2020 (COVID-19 crash)
  - all four are skewed to the left and heavy-tailed

# The Normal Distribution Does Not Fit



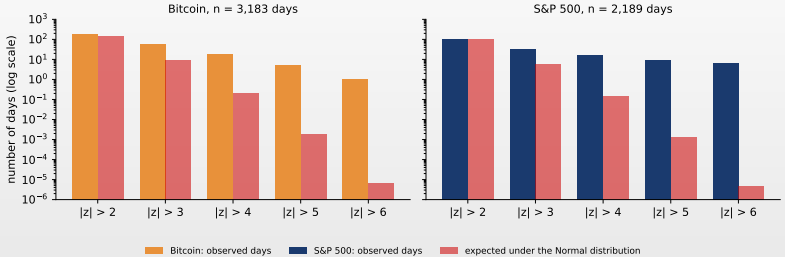

- Left: too many quiet days in the centre and too many extreme days in the tails, too few in between
- **QQ plot**: sorted returns against the quantiles a Normal distribution would give; Normal data lie on the red line
- Both QQ plots bend away at the ends: heavy tails in crypto *and* in equities

BA_L2_hist_qq ▶ Interactive

## How Rare Is a Crash?

- Standardise: $z_t = (r_t - \bar{r})/s$, the number of standard deviations of each day
- Under the Normal distribution
  - $P(|z| > 3) = 0.27\%$; $P(|z| > 4) = 0.0063\%$
  - in 3 183 days we would expect 0.2 days beyond 4 standard deviations
- Bitcoin, 2018–2026: **17 days** beyond 4 standard deviations
- Worst day, 12 March 2020: log return $-46.5\%$, i.e. $z = -13.9$
  - as a simple return: $e^{-0.465} - 1 = -37.2\%$ of the value held
  - under the Normal: $P(z \leq -13.9) \approx 5 \times 10^{-44}$, i.e. once in $2 \times 10^{43}$ days
- Business lesson: a risk model built on the Normal distribution will be surprised many times; Lecture 4 fixes this

# Crash Days: Observed against the Normal Distribution



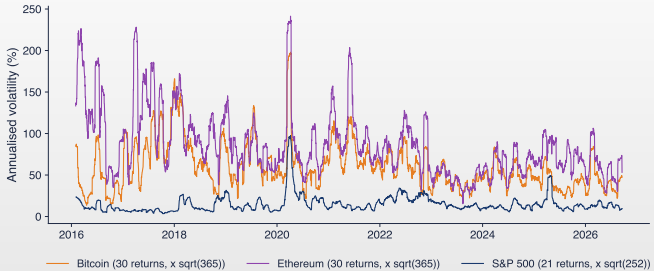

- Days with $|z_t| > k$, $k = 2, \ldots, 6$, in 2018–2026, against $n \times P(|z| > k)$ under the Normal distribution (log scale)
- $k = 2$: Bitcoin 185 days against 145 expected, S&P 500 99 against 100: the Normal fits the moderate days
- $k = 4$: Bitcoin 17 against 0.20 (84 times as many), S&P 500 15 against 0.14 (108 times)
- $k = 6$: Bitcoin 1 day, S&P 500 6 days; the Normal expects one such day in 507 million
- The ratio of observed to expected grows with $k$: the tails, not the centre, break the Normal distribution

BA_L2_crash_counts

## Worked Example: Expected Crash Days under the Normal

- Bitcoin, 2018–2026: $n = 3\,183$ daily returns
- Step 1, $k = 3$: $P(|z| > 3) = 2 \times 0.00135 = 0.00270$
  - expected days $= 3\,183 \times 0.00270 = 8.6$; observed 59, seven times as many
- Step 2, $k = 4$: $P(|z| > 4) = 2 \times 3.17 \times 10^{-5} = 6.33 \times 10^{-5}$
  - expected days $= 3\,183 \times 6.33 \times 10^{-5} = 0.20$; observed 17, 84 times as many
- Step 3, waiting time between two days with $|z| > 4$
  - under the Normal: $1/(6.33 \times 10^{-5}) = 15\,787$ days, about 43 years
  - observed: $3\,183/17 = 187$ days, about six months
- Reading: a risk limit built on Normal tail probabilities treats a once-in-six-months event as a once-in-43-years event

# Volatility Through Time




- Volatility on about one month: 30 daily returns $\times\sqrt{365}$ for crypto, 21 returns $\times\sqrt{252}$ for the S&P 500
- Medians: 54% (Bitcoin), 77% (Ethereum), 12% (S&P 500); all three peak in March–April 2020
- Bitcoin's volatility trends down, but it was below that of the S&P 500 on only 0.1% of the days

SFM_ch14_asset_class ▶ Interactive

## Autocorrelation and the Ljung–Box Test

- **Sample autocorrelation** at lag $k$:

$$\hat{\rho}(k) = \frac{\sum_{t=k+1}^{n}(r_t - \bar{r})(r_{t-k} - \bar{r})}{\sum_{t=1}^{n}(r_t - \bar{r})^2}$$

  - under independence, $\hat{\rho}(k)$ lies within $\pm 1.96/\sqrt{n}$ with probability 95%
  - autocorrelation function (ACF) of $r_t$: is the direction predictable? ACF of $r_t^2$: is the size of the move predictable?

- **Ljung–Box** test of the first $m$ autocorrelations together:

$$Q(m) = n(n+2)\sum_{k=1}^{m}\frac{\hat{\rho}(k)^2}{n-k} \sim \chi^2_m \text{ under } H_0\text{: no autocorrelation}$$

  - Bitcoin daily log returns, 18 September 2014 – 18 September 2026, $n = 4\,384$, $m = 10$ lags
  - $Q = 20.2$ (p-value 0.028) for $r_t$; $Q = 212.7$ for $r_t^2$; the 5% critical value of $\chi^2_{10}$ is 18.3
  - weak dependence in returns, strong dependence in squared returns

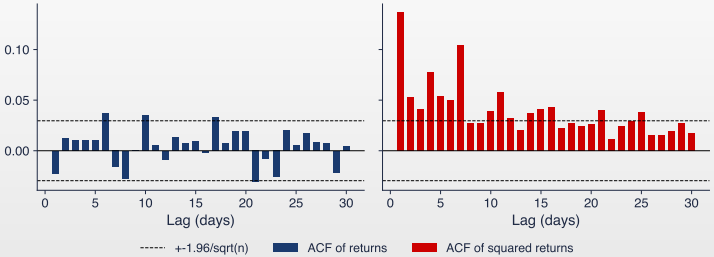
## Worked Example: Ljung–Box with Two Lags

- Bitcoin daily log returns since 2014, $n = 4\,384$; $Q(2) = n(n+2)\left[\hat{\rho}(1)^2/(n-1) + \hat{\rho}(2)^2/(n-2)\right]$
- Returns: $\hat{\rho}(1) = -0.0223$, $\hat{\rho}(2) = 0.0121$

$$Q(2) = 4\,384 \times 4\,386 \times \left(\frac{0.000498}{4\,383} + \frac{0.000147}{4\,382}\right) = 2.83$$

  - below 5.99, the 5% critical value of $\chi^2_2$ (p-value 0.24): we do not reject "no autocorrelation"
- Squared returns: $\hat{\rho}(1) = 0.1367$, $\hat{\rho}(2) = 0.0527$, so $Q(2) = 94.2$ (p-value $4 \times 10^{-21}$): rejected
  - the first term alone, $\approx n\,\hat{\rho}(1)^2 = 4\,384 \times 0.0187 = 81.9$, decides the test
- One lag at a time: the 95% band is $\pm 1.96/\sqrt{4\,384} = \pm 0.030$; $\hat{\rho}(1)$ of returns lies inside it, that of squared returns far outside

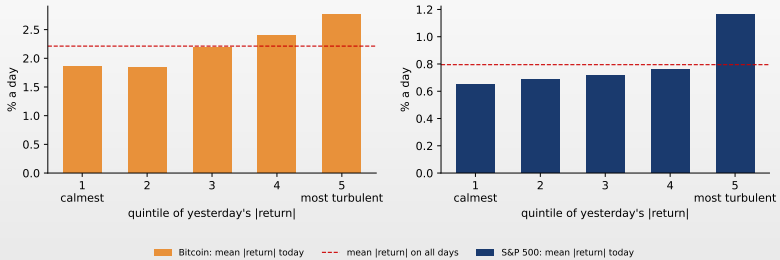
# Volatility Clusters




- **Autocorrelation** at lag $k$: the correlation between today's value and the value $k$ days ago; dashed: the 95% band of pure noise
- Returns (left): almost no autocorrelation; the *direction* of tomorrow's move is hard to predict
- Squared returns (right): clearly positive; the *size* of tomorrow's move is predictable: calm days follow calm days, turbulent days follow turbulent days
- This is **volatility clustering**, the basis of the GARCH models of Lecture 4

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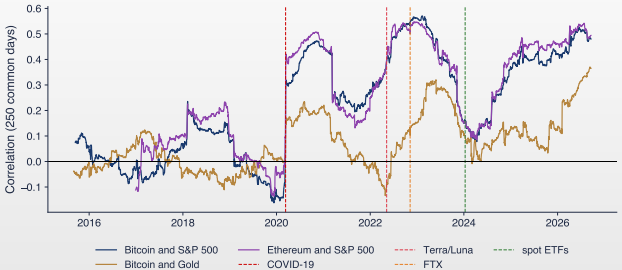
# Calm Follows Calm




- Days of 2018–2026 sorted into five equal groups (quintiles) by yesterday's absolute return; bars: today's mean absolute return
- Bitcoin: 1.86% after the calmest days, 2.77% after the most turbulent, 1.49 times as much; all days 2.21%
- S&P 500: 0.65% and 1.17%, 1.80 times as much; all days 0.79%
- Yesterday's size says something about today's size, not about its sign: the volatility clustering seen in the ACF of $r_t^2$

BA_L2_calm ▶ Interactive

# Does Bitcoin Move with Equities?




- Correlation of daily returns over the last 250 common trading days, about one year (method: Appendix)
- Before 2020: no link with the S&P 500 (0.02); after: 0.39 on average, peak 0.57 in February 2023
- Since 2020 Bitcoin co-moves with equities, as a **risk asset** would; in 2022 both fell while interest rates rose

SFM_ch14_correlation ▶ Interactive

## Weak-Form Efficiency and the Variance Ratio

- Fama (1970): in a **weak-form efficient** market, past prices do not help to predict future returns
  - if they did, traders would exploit the pattern until it disappeared
- **Variance ratio** (VR) Lo & MacKinlay (1988): compare the variance of $q$-day returns with $q$ times the daily variance

$$\mathrm{VR}(q) = \frac{\mathrm{Var}(r_t + r_{t-1} + \cdots + r_{t-q+1})}{q\,\mathrm{Var}(r_t)}$$

  - random walk (no predictability): $\mathrm{VR} = 1$
  - $\mathrm{VR} > 1$: trends (momentum); $\mathrm{VR} < 1$: reversals (mean reversion)
  - a test statistic $Z^*$, robust to volatility clustering, rejects $\mathrm{VR}(q) = 1$ at 5% if $|Z^*| > 1.96$
- The VR test checks one implication of weak-form efficiency: no linear autocorrelation of returns up to horizon $q$

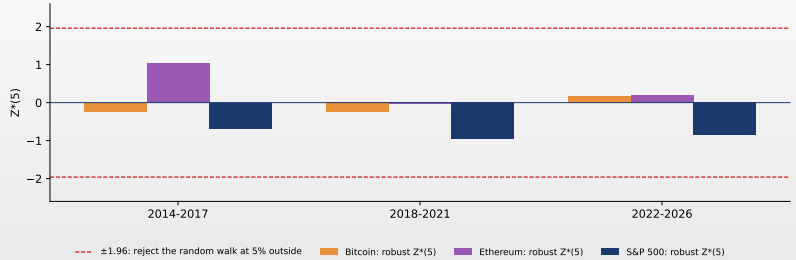
## Variance-Ratio Tests

| | VR(2) | VR(5) | VR(10) | $Z^*(5)$ (p-value) |
|---|---|---|---|---|
| Bitcoin | 0.977 | 0.991 | 1.027 | −0.19 (0.853) |
| Ethereum | 0.988 | 1.067 | 1.137 | 1.17 (0.241) |
| S&P 500 | 0.875 | 0.849 | 0.813 | −1.28 (0.201) |

- Daily log returns since 2014 (Ethereum since 2016) to 18 September 2026
- Interpretation
  - all variance ratios are close to 1; no test rejects $\mathrm{VR}(5) = 1$ at 5%
  - the same holds in each sub-period 2014–2017, 2018–2021, 2022–2026
  - we do not detect the tested deviation from a random walk at these horizons; this does not prove efficiency
  - profitability after trading costs and risk adjustment is a separate question

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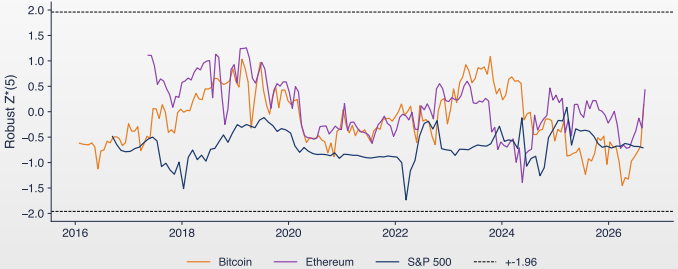
# Variance Ratios by Sub-Period




- Robust $Z^*(5)$ in three sub-periods, to 18 September 2026 (Ethereum from 2016); dashed: $\pm 1.96$
- Bitcoin $-0.23$, $-0.25$, $+0.18$; Ethereum $+1.03$, $-0.04$, $+0.20$; S&P 500 $-0.68$, $-0.95$, $-0.85$
- No bar crosses $\pm 1.96$: we do not reject the random walk in any sub-period
- S&P 500: $\mathrm{VR}(5)$ is 0.92, 0.78 and 0.92, below 1 each time; mild mean reversion of an index, not significant

BA_L2_vr_periods ▶ Interactive

# Efficiency Through Time




- Robust $Z^*(5)$ on windows of 500 days, one window every 21 days; dashed: $\pm 1.96$
- No 500-day window rejects $\mathrm{VR}(5) = 1$ for Bitcoin at 5%
- Early studies found Bitcoin inefficient before 2013–2016 Urquhart (2016); markets became more efficient as they grew Tran & Leirvik (2020)

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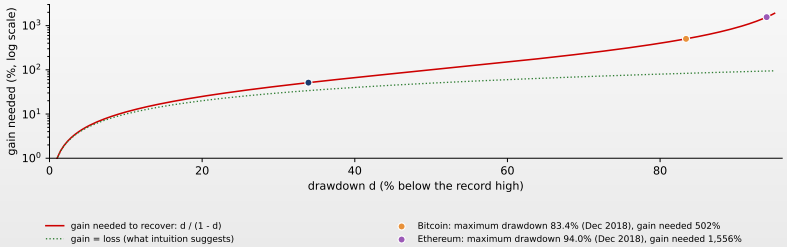
## What Efficiency Does Not Mean

- Not rejecting $\mathrm{VR} = 1$: no detectable linear predictability of returns from past returns at these horizons
  - other patterns (nonlinear, or conditional on volume or news) are not tested
  - even full weak-form efficiency says nothing about whether the price is *right*
- A price can be unpredictable and still far from any fundamental value
  - Bitcoin has no cash flows: no dividends, no profits to discount
  - so "fundamental value" must come from use: payments, settlement, collateral, network size
- Two questions remain
  - how far can the price fall from a peak? (drawdowns)
  - can we detect a bubble, and does network activity justify the price? (log-periodic power law (LPPL), Metcalfe)

## Drawdown: How Far from the Peak

- **Drawdown** on day $t$: $D_t = P_t / \max_{s \le t} P_s - 1$, the loss from the last record high
  - **maximum drawdown**: the deepest fall; **time under water**: the days until a new record
- A loss of $d$ needs a gain of $d/(1-d)$ to recover
  - $d = 50\%$: $+100\%$; $d = 75\%$: $+300\%$; $d = 83.4\%$: $+502\%$
- Why managers care: a firm holding crypto on its balance sheet reports the drawdown, not the average return

# The Gain Needed to Recover



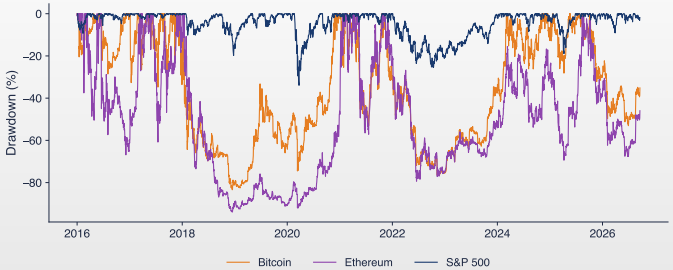

- Red: the gain $d/(1-d)$ that recovers a drawdown $d$, log scale; dotted green: a gain equal to the loss
- Maximum drawdowns of the daily closes since 2016
  - S&P 500: 33.9% (March 2020), recovered by a gain of 51%
  - Bitcoin: 83.4% (December 2018), gain needed 502%; Ethereum: 94.0% (December 2018), 1 556%
- Every extra point of loss needs more than one point of gain; beyond a loss of 50% the gain needed exceeds 100%

BA_L2_recovery

## Worked Example: a Drawdown and Its Recovery

- Bitcoin: peak on 8 November 2021 at 67 567 USD; trough on 21 November 2022 at 15 787 USD
- Step 1, depth: $d = 1 - 15\,787/67\,567 = 76.6\%$
- Step 2, gain needed: $d/(1-d) = 0.766/0.234 = 328\%$, the same as $67\,567/15\,787 - 1$
- Step 3, time: new record on 4 March 2024, 847 days after the peak and 469 days after the trough
- Step 4, speed of the recovery: $4.28^{1/1.28} - 1 = 210\%$ a year from the trough
  - at a steady 50% a year the same recovery would take $\ln 4.28 / \ln 1.5 = 3.6$ years
- Reading: the time under water depends on how fast prices rise after the trough, which no one knows at the trough

# Drawdowns of Bitcoin, Ethereum and the S&P 500




- Daily closes since 2016; maximum drawdowns: Bitcoin −83.4%, Ethereum −94.0%, S&P 500 −33.9%
- Bitcoin spent 76% of the days more than 10% below its record; the S&P 500 16%

SFM_ch14_bubbles_etf ▶ Interactive

# Bubbles and Crashes

- **Bubble**: a price far above any fundamental value, sustained by the expectation of selling higher
  - Cheah & Fry (2015): the price of Bitcoin shows speculative bubbles
- The idea of Johansen, Ledoit & Sornette (2000): imitation between traders creates **positive feedback**
  - prices grow *faster than exponentially*: the growth rate itself grows
  - the crash hazard rises as the price approaches a **critical time** $t_c$
  - the hierarchy of traders (individuals, funds, markets) adds oscillations that speed up near $t_c$
- The result is the **log-periodic power law** (LPPL), next slide



Didier Sornette, co-author of the LPPL model

Photo: Didier Sornette (2012); CC BY-SA 3.0 DE; Wikimedia Commons

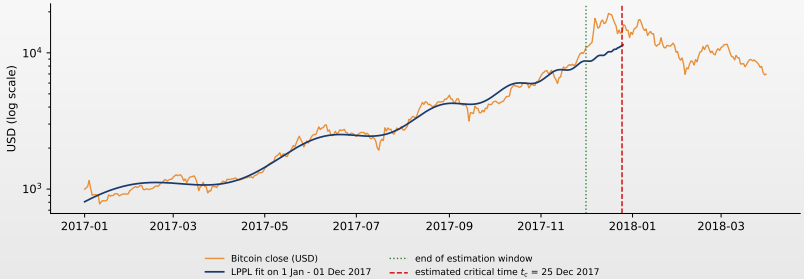
## The LPPL Model

- Log price before the critical time, $t < t_c$; the model describes $\mathrm{E}[\ln P_t]$, and $\varepsilon_t$ is the error:

$$\ln P_t = A + B\,(t_c - t)^m + C\,(t_c - t)^m \cos\big(\omega \ln(t_c - t) - \phi\big) + \varepsilon_t$$

- Parameters
  - $t_c$: critical time, the end of the bubble regime implied by the fitted model, not a guaranteed crash date; $A$: the log price reached at $t_c$
  - $B < 0$ and $0 < m < 1$: super-exponential growth; $(t_c - t)^m$ shrinks to 0 as $t \to t_c$
  - $C$: amplitude of the oscillations; $\omega$: their log-frequency; $\phi$: their phase
- **Log-periodic**: the peaks of the cosine come at times $t_k$ with $(t_c - t_{k+1})/(t_c - t_k) = e^{-2\pi/\omega}$
  - the gaps between peaks shrink geometrically: the oscillations accelerate towards $t_c$
- Constraints in the estimation: $0.1 \le m \le 0.9$, $6 \le \omega \le 13$, $t_c$ after the last observation
- Bubble filter on the fitted parameters: $\hat{B} < 0$ and $|\hat{C}|/|\hat{B}| < 1$

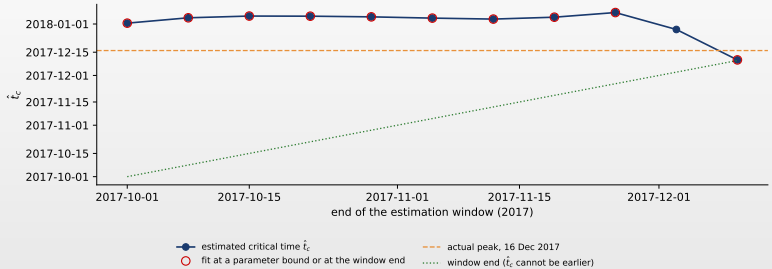
# Bitcoin 2017: an LPPL Fit




- Daily log close, window 1 January – 1 December 2017 ($n = 335$); two-step estimation, least squares inside a search over $(t_c, m, \omega)$ (Appendix)
- Estimated $t_c$ = 25 December 2017; the actual peak came on 16 December 2017 (19 497 USD), followed by a fall of 83%
- $R^2 = 0.98$ on a window chosen after the crash: a good retrospective fit, not a dated forecast (next slide)

BA_L2_lppl_btc2017 ▶ Interactive

# $\hat{t}_c$ Window by Window




- Eleven windows, all starting 1 January 2017, ending weekly from 1 October to 10 December 2017
- Nine of eleven fits sit at the bounds $m = 0.9$, $\omega = 13$ (red circles): the parameters are poorly identified; their $\hat{t}_c$ lie between 1 and 7 January 2018
- Only the window ending 3 December gives an interior fit ($\hat{t}_c$ = 28 December 2017)
- Window ending 10 December: $\hat{t}_c$ equals the last observation, a degenerate solution

BA_L2_lppl_tc ▶ Interactive

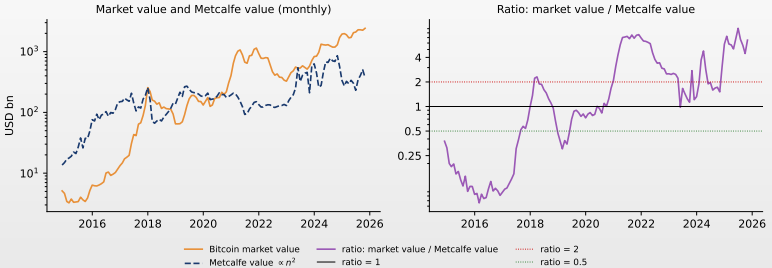
## Network Value: Metcalfe's Law

- **Metcalfe's law**: the value of a network grows with the square of its users $n$
  - $n$ users can form about $n^2/2$ connections; each connection adds value
  - used to value telecom and social networks; applied to Bitcoin by Peterson (2018)
- As a regression on logs:

$$\ln M_t = a + b \ln n_t + u_t, \qquad \text{Metcalfe: } b = 2$$

  - $M_t$: market value; $n_t$: a measure of use, here Bitcoin transactions per day; $u_t$: the error
  - $b$ is an **elasticity**: 1% more use goes with $b$% more value
- Data: monthly means, November 2014 – October 2025, 132 months; the 11 erroneous days of 2023 (Lecture 1) excluded
- Limits, before any chart
  - transactions are not users: one firm can send many transactions; exchange users trade off-chain
  - two series with trends can give a high $R^2$ without a stable link (spurious regression)

# Market Value against Metcalfe Value




- Metcalfe value: $\exp(\hat{c} + 2 \ln n_t)$, with $b$ fixed at 2 and $\hat{c}$ set so that the ratio is 1 on average in logs
- The ratio was below 0.5 in 2015–2016 and above 4 in 2021 and 2025 (maximum 9.1, June 2025)
- The ratio drifts upwards: in this sample value grew faster than transactions
- The ratio is a diagnostic that depends on the model ($b = 2$) and on the proxy (transactions), not a fair value

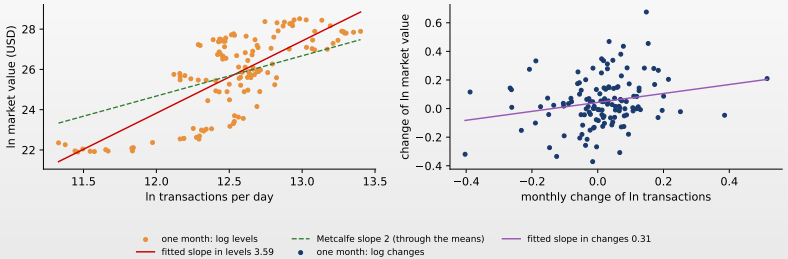
BA_L2_metcalfe ▶ Interactive

# Metcalfe's Law: the Estimates

| Sample | months | $\hat{b}$ | standard error | $R^2$ |
|---|---|---|---|---|
| 2014–2025 | 132 | 3.59 | 0.43 | 0.58 |
| 2014–2019 | 62 | 2.71 | 0.35 | 0.52 |
| 2020–2025 | 70 | 1.17 | 0.48 | 0.20 |
| 2014–2025, with a linear time trend | 132 | 0.27 | 0.47 | 0.89 |
| 2014–2025, monthly changes $\Delta \ln M_t$ on $\Delta \ln n_t$ | 131 | 0.31 | 0.15 | 0.05 |

- Newey–West standard errors (12 lags in levels, 3 in changes)
- Levels, full sample: test of $b = 2$, $t = (3.59 - 2)/0.43 = 3.66$; rejected at 5%
- But the levels regression may be spurious: Engle–Granger test, no cointegration (p-value 0.57)
  - with a time trend or in monthly changes, $\hat{b}$ falls to about 0.3
- Since 2020 the link is weak ($R^2 = 0.20$); ETFs, institutional flows and macro news are plausible drivers, not identified here

# Metcalfe's Law in Levels and in Changes




- Left: log levels, 132 months; fitted slope 3.59, against Metcalfe's slope of 2 (dashed green)
- Right: monthly log changes, 131 months; slope 0.31 (SE 0.15), $R^2 = 0.05$, correlation 0.22
- Both levels trend upwards, so their cloud is long and tilted; changes remove the trends, and the link almost disappears
- A relation that holds in levels but not in changes is the warning sign of a spurious regression

BA_L2_metcalfe_changes

## Summary

- Annualise with the market's own calendar: $\sigma_a = s\sqrt{365}$ gives 64.1% for Bitcoin, against 19.2% for the S&P 500
- Normality is rejected (Jarque–Bera $> 10\,000$ for all four assets); crash days are far more frequent than the Normal distribution allows
- Volatility clusters: Ljung–Box $Q(10) = 212.7$ for $r_t^2$, against 20.2 for $r_t$
- Since 2020 Bitcoin co-moves with equities, like a risk asset: correlation 0.39 with the S&P 500
- Robust variance-ratio tests do not reject a random walk at 2–10 days, in any sub-period; this is not proof of efficiency
- LPPL fits the 2017 bubble with $R^2 \approx 0.98$, but $\hat{t}_c$ moves with the window: a good in-sample fit does not date the crash
- Metcalfe in levels: $\hat{b} = 3.59$, standard error (SE) 0.43; no cointegration, and in monthly changes $\hat{b} = 0.31$
- Lecture 4 models these facts: volatility clustering (GARCH) and heavy tails (VaR 1%, ES 2.5%)

## Interpretation Exercises

1. A treasurer reports Bitcoin's annual volatility as 53.3%. What did she do wrong, and what is the right figure?
2. Jarque–Bera $= 29\,620$ for Bitcoin. State $H_0$, the conclusion, and the consequence for a VaR computed with the Normal distribution.
3. Bitcoin, 2014–2026: Ljung–Box $Q(10) = 20.2$ (p 0.028) for $r_t$, 212.7 for $r_t^2$; robust $Z^*(5) = -0.19$ (p 0.853). Is Bitcoin predictable? In what sense? What do the tests not cover?
4. LPPL fit on 1 January – 1 December 2017: $R^2 = 0.98$, $\hat{t}_c =$ 25 December 2017. Would you have sold on 1 December? What does a high $R^2$ not tell you?
5. Metcalfe: $\hat{b} = 3.59$ (SE 0.43) in levels on 2014–2025, no cointegration (p 0.57), $\hat{b} = 0.31$ (SE 0.15) in monthly changes. What does this imply for valuations based on user growth?

Watch Robust Markowitz for Crypto-Assets

# Appendix: Further Material

Optional material for self-study.

- The Tail Index and the Hill Estimator
- Tail Indices of Crypto, Equities and Gold
- Tail Plot and Hill Plot
- Is Kurtosis a Stable Number?
- A 7-Day Market with a Weekend Lull
- Three Crisis Episodes
- Hedge, Safe Haven or Risk Asset?
- Average Returns on the Worst Equity Days
- Estimating the LPPL
- LPPL Estimates: How Stable Is $\hat{t}_c$?
- A Rolling Metcalfe Elasticity
- Measuring Volatility Through Time
- Rolling Correlation
- The Robust Variance-Ratio Statistic
- Worked Example: Variance Ratio from Autocorrelations
- The Variance-Ratio Profile
- Bitcoin's Large Drawdowns
- Worked Example: Accelerating Oscillations

## The Tail Index and the Hill Estimator

- Far in the tail, losses follow a power law:

$$P(L > x) \approx c\, x^{-\alpha} \quad \text{for large } x$$

  - $L = -r$: the daily loss; $c > 0$: a constant; $\alpha$: the **tail index**
  - smaller $\alpha$ = heavier tail; moments of order $q \geq \alpha$ are infinite: $\alpha < 4$ means infinite kurtosis
- Hill (1975) estimator from the $k$ largest losses $L_{(1)} \geq L_{(2)} \geq \cdots \geq L_{(k+1)}$, with a confidence interval (CI):

$$\hat{\alpha} = \left[ \frac{1}{k} \sum_{i=1}^{k} \ln \frac{L_{(i)}}{L_{(k+1)}} \right]^{-1}, \qquad \mathrm{CI}_{95\%} = \hat{\alpha}\left(1 \pm \frac{1.96}{\sqrt{k}}\right)$$

  - $L_{(k+1)}$: the threshold of the tail; here $k = 2.5\%$ of $n$
  - large ratios $L_{(i)}/L_{(k+1)}$ on average give a small $\hat{\alpha}$: a heavy tail

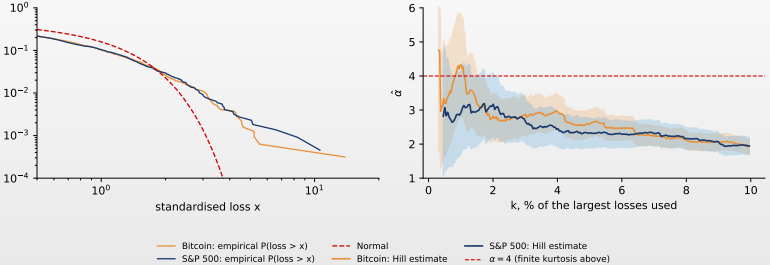
# Tail Indices of Crypto, Equities and Gold

| | left tail $\hat{\alpha}$ | 95% CI | right tail $\hat{\alpha}$ | 95% CI |
|---|---|---|---|---|
| Bitcoin | 2.71 | [2.20, 3.21] | 3.37 | [2.74, 4.00] |
| Ethereum | 2.62 | [2.10, 3.15] | 3.09 | [2.48, 3.71] |
| S&P 500 | 2.68 | [2.08, 3.29] | 2.80 | [2.17, 3.43] |
| Gold | 3.11 | [2.42, 3.80] | 3.04 | [2.36, 3.71] |

- Daily returns since 2014 (Bitcoin, S&P 500, gold) and 2016 (Ethereum), to 18 September 2026
- Interpretation
  - all left-tail indices are below 4: the sample kurtosis of the four-asset table is not a stable number (next slides)
  - crypto is more *volatile*, but the *shape* of its tails is close to that of equities; the intervals overlap
  - for Bitcoin, losses have a heavier tail than gains (2.71 against 3.37)

SFM_ch14_asset_class

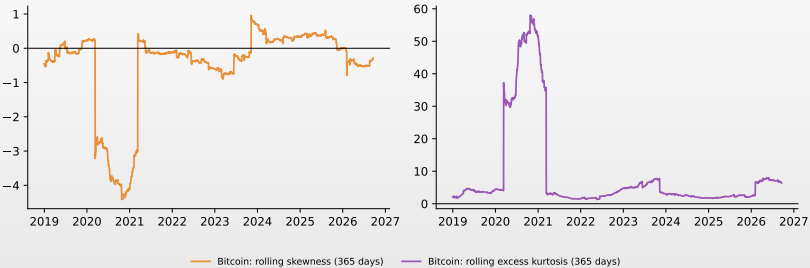
# Tail Plot and Hill Plot




- Left: $P(\text{loss} > x)$ on log-log scales; a power-law tail is a straight line with slope $-\alpha$, the Normal (red) bends down
- Right: Hill estimate against $k$ with 95% bands, 2018–2026. Bitcoin: 4.33 ($k = 1\%$), 2.71 (2.5%), 2.60 (5%), 1.94 (10%); S&P 500: 2.92, 2.81, 2.32, 1.95
- Small $k$: few observations, wide bands; large $k$: the centre of the distribution biases $\hat{\alpha}$ down. Report $\hat{\alpha}$ on the flat part, here $k \approx 2.5\%$

BA_L2_tails

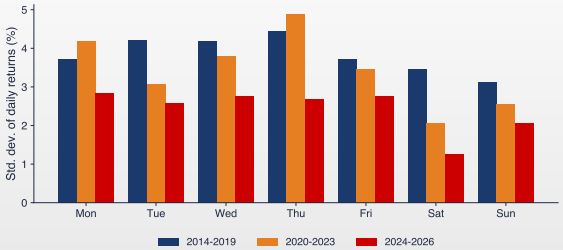
# Is Kurtosis a Stable Number?




- Skewness and excess kurtosis of Bitcoin on rolling 365-day windows
- Kurtosis ranges from 1.3 to 58.0 (window ending October 2020, which contains 12 March 2020); today 6.4
- One day moves the estimate by a factor of ten: with $\alpha < 4$ the sample kurtosis has no stable limit (previous slides)

BA_L2_rolling_moments

# A 7-Day Market with a Weekend Lull




- Standard deviation of Bitcoin's daily returns by closing day, three periods
- No weekend effect in the *mean*; but weekend variance is only 35–66% of weekday variance
- Crypto trades on Saturdays, but banks, macro news and institutional traders do not: risk follows the traditional calendar

SFM_ch14_asset_class

# Three Crisis Episodes

| Episode | period | Bitcoin | Ethereum | S&P 500 | gold |
|---|---|---|---|---|---|
| COVID-19 | 19 Feb – 23 Mar 2020 | −40.6 | −65.5 | −41.4 | 0.3 |
| Terra/Luna | 5 – 12 May 2022 | −23.0 | −33.8 | −5.4 | −3.0 |
| FTX | 5 – 21 Nov 2022 | −29.9 | −38.4 | 4.6 | 3.4 |

- Cumulative log returns over each episode, in log-return percentage points
  - as simple returns $e^{r/100} - 1$: COVID-19 Bitcoin −33.4%, Ethereum −48.1%, S&P 500 −33.9%
- Interpretation
  - a global shock (COVID-19) hits crypto as hard as equities; gold does not move
  - crypto-specific shocks (the collapse of the stablecoin TerraUSD; the bankruptcy of the exchange FTX) cost Bitcoin and Ethereum at least 20% (simple returns), equities hardly anything
  - in these three episodes the link is **asymmetric**: the equity crash reached crypto, the crypto crashes did not reach equities

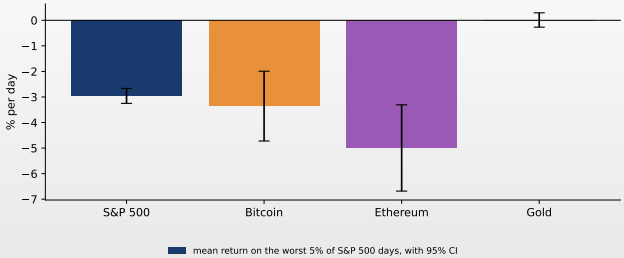
## Hedge, Safe Haven or Risk Asset?

- Baur & Lucey (2010): a **hedge** is uncorrelated or negatively correlated with equities on average; a **safe haven** is uncorrelated or negatively correlated with equities in extreme market stress
- A simpler check here: the mean return of asset $x$ on the set $\mathcal{S}$ of the worst 5% of S&P 500 days

$$\bar{x}_{\mathcal{S}} = \frac{1}{n_{\mathcal{S}}} \sum_{t \in \mathcal{S}} x_t, \qquad t = \frac{\bar{x}_{\mathcal{S}}}{s_{\mathcal{S}} / \sqrt{n_{\mathcal{S}}}}$$

  - 2018–2026, common days: $n_{\mathcal{S}} = 110$ days with the S&P 500 below $-1.77\%$, on average $-2.96\%$
- Means with 95% CI: Bitcoin $-3.36\%$ $[-4.72, -1.99]$, Ethereum $-4.99\%$ $[-6.68, -3.31]$, gold $+0.01\%$ $[-0.27, 0.29]$
- Reading
  - on these days crypto falls on average more than equities
  - gold: a zero mean is not rejected; this does not show that its mean is non-negative, nor that gold is a safe haven in the sense of Baur & Lucey (2010)

# Average Returns on the Worst Equity Days




- Mean daily return on the 110 worst S&P 500 days of 2018–2026, with 95% confidence intervals
- For a corporate treasury: in this sample, Bitcoin lost on average on the days when equities were under stress

BA_L2_safe_haven

## Estimating the LPPL

- Rewrite the cosine term with $C_1 = C\cos\phi$ and $C_2 = C\sin\phi$:

$$\ln P_t = A + Bf_t + C_1 g_t + C_2 h_t + \varepsilon_t$$

$$f_t = (t_c - t)^m, \quad g_t = (t_c - t)^m \cos(\omega \ln(t_c - t)), \quad h_t = (t_c - t)^m \sin(\omega \ln(t_c - t))$$

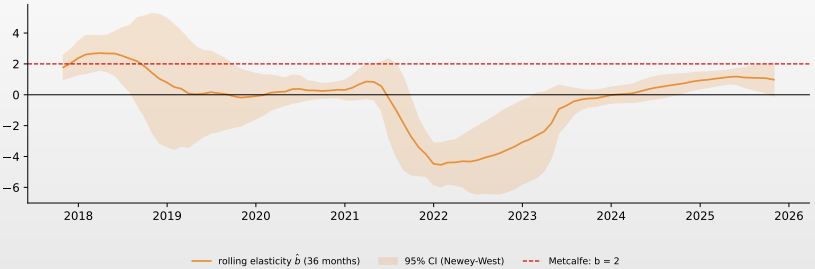
- Two steps
  1. for given $(t_c, m, \omega)$, the model is linear in $(A, B, C_1, C_2)$: ordinary least squares
  2. search over $(t_c, m, \omega)$ to minimise the sum of squared errors $\sum_t \varepsilon_t^2$
- Caution
  - many local minima: start the search from many points; fit on many windows
  - $t_c$ is the least precise parameter; fits look convincing after the crash, real-time warnings give false alarms

# LPPL Estimates: How Stable Is $\hat{t}_c$?

| window ends | $n$ | $\hat{t}_c$ | lead (days) | $\hat{m}$ | $\hat{\omega}$ | $\lvert\hat{C}\rvert/\lvert\hat{B}\rvert$ | $R^2$ |
|---|---|---|---|---|---|---|---|
| 15 Nov 2017 | 319 | 4 Jan 2018 | 50 | 0.900 | 13.00 | 0.089 | 0.980 |
| 1 Dec 2017 | 335 | 25 Dec 2017 | 24 | 0.851 | 12.49 | 0.088 | 0.982 |
| 8 Dec 2017 | 342 | 8 Dec 2017 | 0 | 0.741 | 11.14 | 0.081 | 0.979 |

- All windows start on 1 January 2017; $\hat{B} \approx -2.8$ in all three: super-exponential growth
- Interpretation
  - $R^2 \approx 0.98$ in every window: a good fit does not mean a reliable forecast
  - $\hat{t}_c$ moves by four weeks when the window moves by three weeks
  - 15 November: $\hat{m}$ and $\hat{\omega}$ sit on the bounds of the search (0.9 and 13), a sign of a poorly identified fit
  - 8 December: $\hat{t}_c$ equals the last observation, a degenerate solution
- Practice: fit many windows and report the share of windows that pass the filters (the log-periodic power law singularity (LPPLS) confidence indicator)

# A Rolling Metcalfe Elasticity




- $\hat{b}$ on rolling 36-month windows with Newey–West 95% intervals; red: Metcalfe's $b = 2$
- $\hat{b}$ peaks at 2.70 (window ending March 2018), turns negative in 2021–2023 (value fell while transactions grew), ends at 0.98 (October 2025)
- The interval covers 2 in only 28% of the windows: there is no stable network-value elasticity

BA_L2_metcalfe_rolling

## Measuring Volatility Through Time

- **Rolling volatility** on a window of the last $w$ days, annualised:

$$\hat{\sigma}_{a,t} = \sqrt{A}\sqrt{\frac{1}{w-1}\sum_{j=0}^{w-1}\left(r_{t-j} - \bar{r}_t\right)^2}, \qquad \bar{r}_t = \frac{1}{w}\sum_{j=0}^{w-1} r_{t-j}$$

  - $w$: the window length (30 days for crypto, 21 trading days for equities, about one month)
  - $A$: returns per year (365 or 252); the window moves forward one day at a time
- Trade-off in the choice of $w$
  - short window: reacts fast, but noisy; long window: smooth, but late
  - the standard error of $\hat{\sigma}$ is roughly $\hat{\sigma}/\sqrt{2w}$: about 13% of $\hat{\sigma}$ for $w = 30$
- Lecture 4 replaces the window by a model: exponentially weighted moving average (EWMA) or GARCH

## Rolling Correlation

- Pearson correlation of two return series $x$ (Bitcoin) and $y$ (S&P 500) on the last $w$ common days:

$$\hat{\rho}_{xy,t} = \frac{\sum_{j=0}^{w-1}(x_{t-j} - \bar{x}_t)(y_{t-j} - \bar{y}_t)}{\sqrt{\sum_{j=0}^{w-1}(x_{t-j} - \bar{x}_t)^2}\sqrt{\sum_{j=0}^{w-1}(y_{t-j} - \bar{y}_t)^2}}$$

  - $w = 250$ common days, about one year of equity trading
  - standard error $\approx (1 - \rho^2)/\sqrt{w} \approx 0.06$: changes below 0.1 are noise
- Two calendars
  - join *prices* on common days first, then compute returns: Monday's Bitcoin return then covers Friday to Monday, like the equity return
  - closes are not simultaneous (midnight UTC, Coordinated Universal Time, vs 16:00 New York): weekly returns give a check

## The Robust Variance-Ratio Statistic

- Volatility clustering makes the independent and identically distributed (i.i.d.) test too eager to reject; Lo & MacKinlay (1988) give a robust version:

$$Z^*(q) = \frac{\widehat{\mathrm{VR}}(q) - 1}{\sqrt{\hat{\theta}^*(q)}} \;\approx\; N(0,1)$$

$$\hat{\theta}^*(q) = \sum_{j=1}^{q-1} \left[\frac{2(q-j)}{q}\right]^2 \hat{\delta}_j, \qquad \hat{\delta}_j = \frac{\sum_{t=j+1}^{n} (r_t - \bar{r})^2 (r_{t-j} - \bar{r})^2}{\left[\sum_{t=1}^{n} (r_t - \bar{r})^2\right]^2}$$

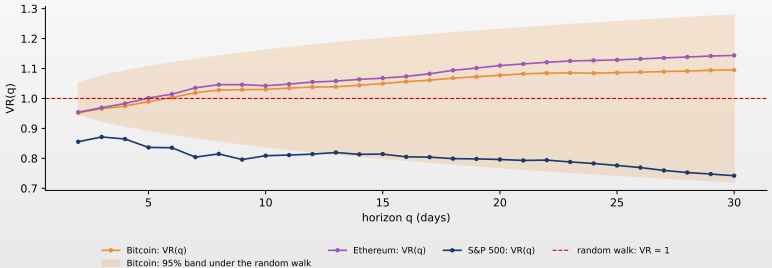
  - $\hat{\delta}_j$: the variance of the lag-$j$ autocorrelation when volatility changes over time
  - reject the random walk at 5% if $|Z^*(q)| > 1.96$

- Link with autocorrelations: $\mathrm{VR}(q) \approx 1 + 2\sum_{j=1}^{q-1} \left(1 - \frac{j}{q}\right)\rho(j)$
  - positive autocorrelations push VR above 1, negative ones below 1

## Worked Example: Variance Ratio from Autocorrelations

- For $q = 2$: $\mathrm{VR}(2) \approx 1 + 2\,(1 - \tfrac{1}{2})\,\rho(1) = 1 + \rho(1)$
- Bitcoin, since 2014: $\hat{\rho}(1) = -0.022$, so $\mathrm{VR}(2) \approx 0.978$; the direct estimate is 0.977
- S&P 500, same period: $\hat{\rho}(1) = -0.126$, so $\mathrm{VR}(2) \approx 0.874$; the direct estimate is 0.875
- Interpretation
  - the S&P 500 index shows mild daily mean reversion, typical of an index (bid-ask bounce, stale prices of its members)
  - the robust test does not reject because the variance of $\widehat{\mathrm{VR}}$ is large when volatility clusters

# The Variance-Ratio Profile




- $\widehat{\mathrm{VR}}(q)$ for $q = 2, \ldots, 30$ days, 2018–2026; shaded: the robust 95% band for Bitcoin
- Bitcoin: VR(5) = 0.989 ($Z^* = -0.19$), VR(20) = 1.077 ($Z^* = 0.66$); Ethereum VR(20) = 1.11 ($Z^* = 0.95$)
- S&P 500: VR(20) = 0.796 ($Z^* = -0.76$): mean reversion in the point estimate, but not significant

BA_L2_vr_profile

# Bitcoin's Large Drawdowns

| Peak | USD | trough | USD | depth (%) | new record | days under water |
|---|---|---|---|---|---|---|
| 2014-09-17 | 457 | 2015-01-14 | 178 | −61.1 | 2015-12-15 | 454 |
| 2017-12-16 | 19 497 | 2018-12-15 | 3 237 | −83.4 | 2020-11-30 | 1 080 |
| 2021-04-13 | 63 503 | 2021-07-20 | 29 807 | −53.1 | 2021-10-19 | 189 |
| 2021-11-08 | 67 567 | 2022-11-21 | 15 787 | −76.6 | 2024-03-04 | 847 |
| 2025-10-06 | 124 753 | 2026-06-30 | 58 559 | −53.1 | not yet | – |

- Episodes since September 2014 deeper than 50%
- Interpretation
  - five falls of more than half in twelve years: a crash of this size is a regular event
  - the 2017 and 2021 peaks took 1 080 and 847 days to recover

SFM_ch14_bubbles_etf

## Worked Example: Accelerating Oscillations

- Peaks of the cosine term satisfy $\omega \ln(t_c - t_k) - \phi = 2\pi k$, so consecutive distances to $t_c$ shrink by a constant factor

$$\frac{t_c - t_{k+1}}{t_c - t_k} = e^{-2\pi/\omega}$$

- Bitcoin 2017 fit (slide "Bitcoin 2017: an LPPL Fit"): $\hat{\omega} = 12.49$, so $e^{-2\pi/12.49} = 0.605$
  - a peak 30 days before $t_c$ is followed by peaks 18.1, 11.0 and 6.6 days before $t_c$
- The ratio $\lambda = e^{2\pi/\omega} = 1.65$ is the "preferred scaling ratio" of the hierarchy of traders in Johansen, Ledoit & Sornette (2000)
- Practical reading: shrinking gaps between local tops are the visual signature of an LPPL bubble